Diptanshu Sikdar

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EDUCATION

University of California San Diego Sep. 2026 – Present
Ph.D. Computer Science

University of California Irvine Sep. 2022 – Jun. 2026
B.S. Computer Science (Intelligent Systems) | GPA: 3.944
Relevant Coursework: Neural Networks & Deep Learning, Machine Learning & Data Mining, Optimization, Data Structures & Algorithms, Project in AI, Project in Bioinformatics, Quantum Computation & Information, Probability in CS, Formal Languages, System Design, Linear Algebra, Discrete Mathematics
Additional Courses: Diffusion Models, Deep Learning, LLM Application Development

RESEARCH EXPERIENCE

Zhang Laboratory at UCI, Irvine, CA – *Undergraduate Research Assistant* Apr. 2023 – present

- Developed an AI-driven prioritization framework that formulates cell repair as a perturbation optimization problem, using **Q-learning reinforcement learning** with **virtual cell models** and an **Agentic AI**-based verification system
- Built a **Mixture-of-Experts vision model** that predicts gene expression from histology images, which achieved **~50% lower error** than state-of-the-art models; also created a **multimodal captioning pipeline** with **~90% accuracy**
- Trained **VAE**s and **GAN**s to synthesize histology images through **disentanglement**, addressing limited data concerns
- Analyzed cell-cell communication in spatial transcriptomics using **statistical** and **clustering** models in **Python** and **R**

PUBLICATIONS / PRESENTATIONS

Sikdar, D., Liu, J., Zhang, J. (2026). CellFix: Genetic Perturbation Optimization for Therapeutic Target Discovery for Crohn's Disease. Poster presented at Undergraduate Research Opportunities Program Symposium at UCI.

Li, X., Hu, Y., Duan, Z., Wang, X., Sun, S. D., **Sikdar, D.**, Liu, Y., Zhang, J. (2026). sMMC-22M: A Context-Aware Dataset and Benchmark for Single-Cell Spatial Transcriptomics. *NeurIPS*, Submitted.

Li, X., Duan, Z., **Sikdar, D.**, Wang, D., Di, X., Liu, Y., & Zhang, J. (2026). BRIDGE: Spatially Aware Bimodal Representation Integration with Hierarchical Graphs for Spatial Transcriptomics Resolution Enhancement. *Bioinformatics*, Submitted.

Sikdar, D., Li, X., & Zhang, J. (2025). CellVision: Generative modeling of tissue morphology based on disentangled biological latent spaces. Poster presented at Undergraduate Research Opportunities Program Symposium at UCI.

Lee, C. Y., Riffle, D., Xiong, Y., Momtaz, N., Lei, Y., Pariser, J. M., **Sikdar, D.**, Hwang, A., Duan, Z., & Zhang, J. (2024). Characterizing dysregulations via cell-cell communications in Alzheimer's brains using single-cell transcriptomes. *BMC Neuroscience, 25*(24).

Sikdar, D., Cui, M., Chau, A., Bhamra, A., Prasanna, S., & McMahan, L. (2023). Hybrid Quantum-Classical Generative Adversarial Network for synthesizing chemically feasible molecules. *Journal of Emerging Investigators*.

Sikdar, D., Clapis, J., & Bhattacharya, S. (2022). Novel Enhanced Unsupervised Quantum Clustering Algorithm. *Journal of Student Research, 11*(2).

WORK EXPERIENCE

MITRE Corp., Bedford, MA – *Network Architecture and Data Analytics Intern* Jun. 2025 – Present

- Developing a **machine learning model** to optimize pipeline task scheduling based on national security priorities
- Optimized **Python**-based graph analytics pipeline with **RustworkX**, improving runtime **by 35%**
- Benchmarked graph analysis libraries (e.g., NetworkX, igraph) and graph databases (e.g., TigerGraph, Neo4j)

Bluebeam, Pasadena, CA – *R&D/Artificial Intelligence Intern* Jun. 2024 – Sep. 2024

- Developed a **Named Entity Recognition (NER)** pipeline using Multimodal Large Language Models to extract construction metadata from architectural drawing PDFs
- Enhanced NER pipeline accuracy from 65% to **89%** using innovative machine learning and custom optimization algorithms (e.g., simulated annealing) for efficient pre- and post-processing

LEADERSHIP EXPERIENCE

Quantum Computing Club at UCI, Irvine, CA

President Jun. 2023 – present

- **Grew membership 10x** to 50 active members and secured **ICS Department affiliation** under Prof. Sandy Irani
- Led **technical workshops** and coding demos to introduce quantum computing to students
- **Organized outreach and cross-club collaboration events** (e.g., high school outreach, workshop with Hack@UCI)
- Invited **guest speakers** from industry (e.g., NVIDIA, Aliro) and academia (e.g., CMU, UCSB, UCI)
- **Recruited** and **mentored** officers, helping them manage club operations and events

Quantum Algorithms Group Lead Sep. 2022 – Jun. 2023

- Led and presented a **research project** at the Symposium on Quantum Undergraduate Inquiry & Discovery 2023

TEACHING AND MENTORSHIP EXPERIENCE

UC Irvine Donald Bren School of Information and Computer Science, Irvine, CA

Reader Sep. 2023 – Dec. 2023

- **Graded** homework, assignments, quizzes, and exams of **683 students** for Boolean Algebra & Logic
- Assisted Graduate TAs by **proofreading** quizzes and midterm exams

Learning Assistant Jan. 2023 – Jun. 2024

- Supported classes in Data Structures & Algorithms, Boolean Algebra & Logic, Python Programming
- **Facilitated** collaborative **discussion sessions** with 10–30 students in class weekly
- Helped students understand algorithms and debug their project code during lab sessions

MIT Beaver Works Summer Institute – *Associate Instructor of Quantum Software course* Jun. 2023 – Aug. 2023

- **Mentored 28 high-school students** on quantum algorithms, coding exercises, and group projects in Q# and Qiskit
- Developed a **Minecraft mod** in Java **from scratch** to model quantum circuits

SELECTED SOFTWARE ENGINEERING PROJECTS

- *Benchmarking Advanced Model Merging Techniques for Low-Rank Adaptation in Stable Diffusion* 2025
  - Implemented LoRA adapter merging techniques (e.g., Linear, DARE-Linear, TIES-SVD) using HuggingFace's PEFT library to improve multi-task image generation
  - Benchmarked merged models using CLIP Score, BRISQUE, and ArtFID to analyze trade-offs between semantic alignment, perceptual quality, and style realism across different adapter combinations
- *PathologyCVAE: Unsupervised Anomaly Detection for Breast Histopathology* 2025
  - Designed, implemented, and evaluated several convolutional VAE architectures to detect cancerous anomalies
  - Achieved ~66% accuracy and 0.70 AUC score despite low-resolution images
- *Portfolio Optimization Using Quadratic Programming Algorithms* 2024
  - Maximized portfolio returns by computing portfolio risk/return trade-offs based on historical stock data via Quadratic Programming algorithms
  - Benchmarked Projected Gradient Descent method with adaptive learning rate versus Ellipsoid Method against CVXPY baselines
- *Docu LLM: An Intelligent Code Utility Recommender using Retrieval-Augmented Language Models* 2023
  - Implemented an intelligent search system that suggests programming libraries based on user's query using Python, Natural Language Toolkit, Google Custom Search API
  - Implemented RAG pipeline to generate summaries of software utilities with GPT3.5 model via LangChain

- *MedBox: A Web Application for On-Demand Prescription Access* 2023
  - Built a full-stack health and prescription assistance app to provide users easy access to medications
  - Developed the complete backend with Python and APImedic while aiding front end design with JavaScript
- *Variational Quantum Convolutional Neural Network to Predict Steering Angles for Self-driving Cars* 2023
  - Developed and tested a Variational Quantum CNN with a classical CNN with a 4-qubit variational circuit
  - Preprocessed data, experimented with various circuit designs, and evaluated them based on accuracy and runtime

TECHNICAL SKILLS

- **Languages**: Python, C++, R, Java
- **Libraries**: Transformers, PEFT, TensorFlow, PyTorch, Scikit-learn, Keras, NumPy, OpenCV, SciPy, LangChain
- **Tools**: Hugging Face, OpenAI API, Slurm, Linux, Conda, LaTeX, Git, GitHub, GitLab

HONORS

- NSF Graduate Research Fellowship 2026
- ICS Project Expo 2nd Place with $125 Prize 2026
- Undergraduate Research Opportunities Program Research Experience Fellowship with $400 funding 2025 – 2026
- Undergraduate Research Opportunities Program Research Experience Fellowship 2024 – 2025
- ICS Honors Program (selected for academic excellence and research potential) 2024 – 2026
- UCI Dean’s Honor List 2022 – 2025

LINKS

- GitHub: https://github.com/dssikdar
- LinkedIn: https://www.linkedin.com/in/diptanshu-sikdar/
- Google Scholar: https://scholar.google.com/citations?user=H-8I8vkAAAAJ&hl=en